

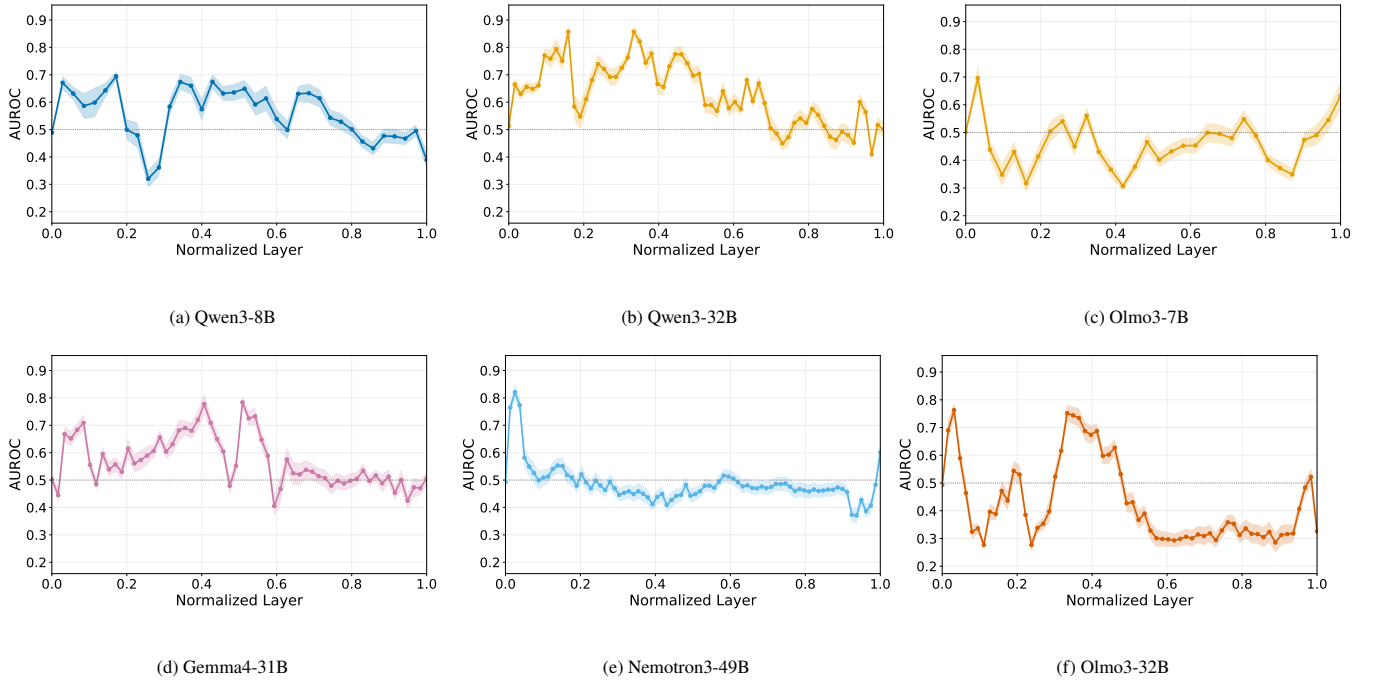


Figure 2: For every model, we report the performance of a linear probe at classifying evaluation vs deployment (measured as AUROC), as a function of the layers (normalized for ease of comparison). The shaded bands indicate $\pm 1\sigma$ across 10 seeds. For every model, the AUROC at the best layer exceeds 0.7, i.e., **for every model there is a layer separating evaluation vs deployment**.

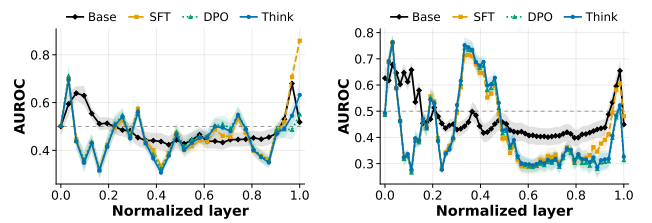
Model	Score 0	Score 1	Score 2	Score 3
Qwen3-8B	94.57	5.27	0.00	0.17
Qwen3-32B	94.73	5.17	0.00	0.10
Olmo3-7B	96.77	3.03	0.00	0.20
Olmo3-32B	93.90	5.97	0.00	0.13
Gemma4-31B	98.70	1.20	0.10	0.00
Nemotron3-49B	99.83	0.10	0.00	0.07

Table 1: Distribution of the judge scores about verbalized evaluation awareness, across six models (3000 samples per model). Score 0 = no awareness; 1 = weak/ambiguous; 2 = explicit awareness; 3 = strategic adaptation. Across all models, more than 93.9% of samples receive score 0.

pairwise cosine similarity of at least 0.998, whereas any of them has a much lower cosine similarity with the base version (0.72 and 0.83 for the 7B and 32B models, respectively). This reinforces the intuition of SFT playing a critical role in shaping the internal representation of evaluation awareness of models.

Verbalization. Observe Table 2: consistently with the classification results, we observe that the verbalization rate increases for both models after the SFT stage. The similarity between the results breaks for the Think stage of the 32B model, where we witness another measurable spike in verbalization rate.

These findings show how verbalization rates tend to increase as the models develop better representations of evaluation awareness—even if, in absolute terms, verbalization



(a) Olmo3-7B AUROC

(b) Olmo3-32B AUROC

Figure 3: Comparison of AUROC across layers for Olmo3-7B (left) and Olmo3-32B (right). Post-trained stages (SFT, DPO, Think) generally cluster together above the Base model, though their ordering is non-monotonic. Normalized layer (normalized layer depth) denotes the layer index divided by the total number of residual stream layers returned by the model.

remains limited in frequency. This prompts a deeper analysis of the correlations between the probe- and the judge-score.

Internal representation and verbalization may not agree locally. The internal and verbalized measures we have introduced agree as average metrics over prompts: models that linearly encode the distinction “evaluation-vs-deployment” also produce some nonzero verbalization, and both signals strengthen after SFT. We next ask whether this agreement extends at the level of individual prompts—that is, whether prompts that the probe scores as evaluation-like are also the

Model	Stage	Score 0	Score 1	Score 2	Score 3
Olmo3-7B	Base	99.50	0.17	0.17	0.17
Olmo3-7B	SFT	96.27	3.57	0.00	0.17
Olmo3-7B	DPO	97.50	2.27	0.00	0.23
Olmo3-7B	Think	96.77	3.03	0.00	0.20
Olmo3-32B	Base	99.10	0.37	0.13	0.40
Olmo3-32B	SFT	96.63	3.30	0.00	0.07
Olmo3-32B	DPO	96.17	3.70	0.00	0.13
Olmo3-32B	Think	93.90	5.97	0.00	0.13

Table 2: Distribution of verbalized evaluation awareness judge scores across Olmo3 7B and 32B training stages.

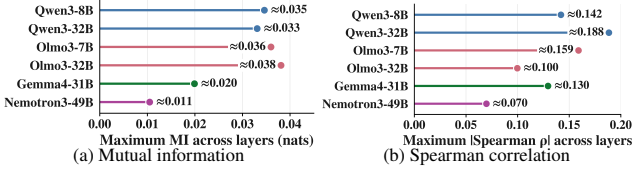


Figure 4: Per-model coupling between the probe score s_ℓ and the judge score J : (a) maximum mutual information $\max_\ell I(s_\ell; J)$ and (b) maximum absolute Spearman correlation $\max_\ell |\rho_\ell|$. Peaks are comparable to the random-direction baseline throughout: on MI, all six models sit within ≈ 0.006 nats of their own baseline peak; on Spearman, four of six fall below it, with only Qwen3-32B (0.188 vs. 0.134) and Olmo3-7B (0.159 vs. 0.124) exceeding it.

ones the judge flags as evaluation-aware. Using the same 1000 MASK prompts and 3 generations per prompt as above, we paired the probe score $s_\ell(x)$ at each layer with the judge score $J(x)$ and computed two complementary statistics: Spearman rank correlation ρ_ℓ (Spearman 1987) and the Kraskov k -nearest-neighbor mutual information $I(s_\ell; J)$ (Kraskov, Stögbauer, and Grassberger 2004); the latter to account for nonlinear dependencies. Notably, across all models and layers, both statistics are very low: peak $|\rho_\ell| < 0.19$ and peak $I(s_\ell; J) < 0.04$ (see Figure 16). The same holds for every Olmo3 checkpoint we tested. In other words, a continuous, above-chance separation of evaluation vs deployment in the activations of a given model does not (necessarily) predict that prompt *verbalizing* being under test. This weak coupling is not an artifact of reading the probe before generation: even at the response-last readout, where the activation and the judged text coincide, both statistics remain at the random-direction baseline. The full results, including all readouts and the Olmo training stages, are in Appendix G.

This dissociation motivates the interventional test in the next section: even if the probe direction does not predict verbalization at a given prompt, *intervening* along the direction of evaluation awareness might change what models verbalize.

4 Activation steering increases verbalization

While not necessarily correlated, are internal representations of evaluation awareness and verbalization causally related? To answer this question, we steer along the probe directions we used for classification (§3) using the protocol outlined

in §2. We employ a dataset X , comprising 1000 prompts from MASK. For every model, we sweep over a subset of its layers and over a maximum range of steering strengths $\alpha \in \{-1.0, -0.8, \dots, 0.8, 1.0\}$ in increments of 0.2; the exact values of α are specified in Figures 5, and 6. For every choice of α and ℓ we calculate the average judge-score:

$$J(\alpha, \ell) := \frac{1}{|X|} \sum_{x \in X} J_{\alpha, \ell}(x).$$

Where $J(0, \ell)$ is the unsteered baseline, we set $\Delta J(\alpha, \ell) := J(\alpha, \ell) - J(0, \ell)$.

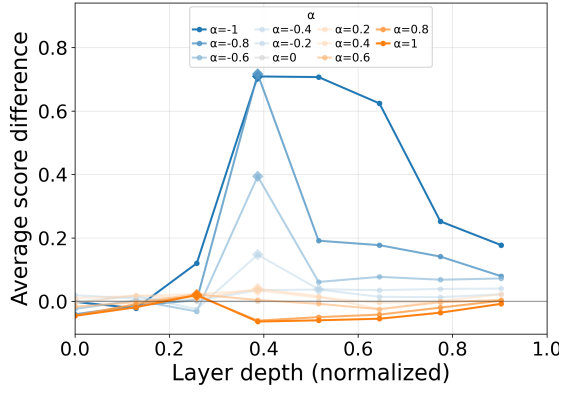
Due to computational constraints, this section focuses only on four models (plus checkpoints), as opposed to six. In particular, we abstain from experimenting with Gemma4-31B and Nemotron3-49B, because (i) the Qwen and Olmo families may reveal patterns of scaling with size, and (ii) the Olmo models can be checked across their checkpoints.

Steering breakdown at large magnitudes. At larger $|\alpha|$, steering can degrade models’ overall performance, rendering our findings less interpretable. For example, we report that Qwen3-32B returns empty responses at certain combinations of steering magnitude and layer. Therefore, these instances are marked with an ‘x’ in the relevant plots and excluded from the average score. The full table of $|\alpha|$ values at which each model first produces empty or degenerate outputs is given in Appendix H.4.

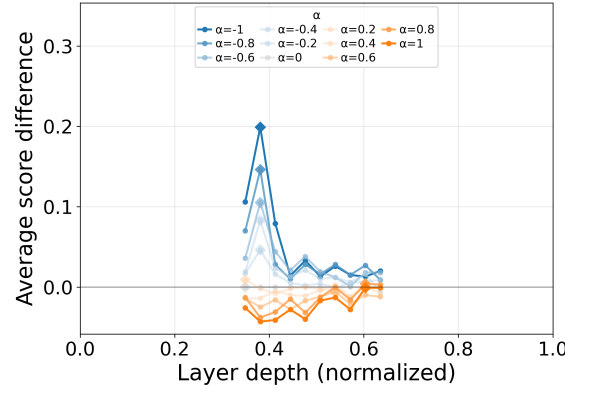
Results. For Olmo3-7B Think, Olmo3-32B Think, and Qwen3-8B, the delta of verbalized responses $\Delta J(\alpha, \ell)$ increases, as predicted, when α takes the negative values (Figure 5). In other words, steering the activations towards the evaluation-like side of $\hat{v}_\ell$ raises the average judge score relative to the unsteered baseline. By contrast, positive values of α produce smaller $\Delta J(\alpha, \ell)$ values, and do not consistently reduce the judge score below baseline. This is expected as the baseline is already very close to zero for most models (Table 10 in Appendix H.3).

The extent to which steering is effective in eliciting verbalization of evaluation awareness is significant, yet model dependent. For Olmo3-7B Think and Olmo3-32B Think, the maximum $\Delta J(\alpha, \ell)$ is approximately 0.7 and 0.2, respectively, at the most evaluation-like α tested. For Qwen3-8B, the peak of $\Delta J(\alpha, \ell)$ is approximately 1.5, with the effect concentrated in a narrow band of layers. For Qwen3-32B, $\Delta J(\alpha, \ell)$ is irregular across layers and includes a substantial number of (α, ℓ) combinations at which raw outputs are empty. We remark that the peak $\Delta J(\alpha, \ell)$ for Qwen3-8B indicates a *strong* verbalization: any score above 1 means that, in expectation, there is some level of verbalization on any prompt.

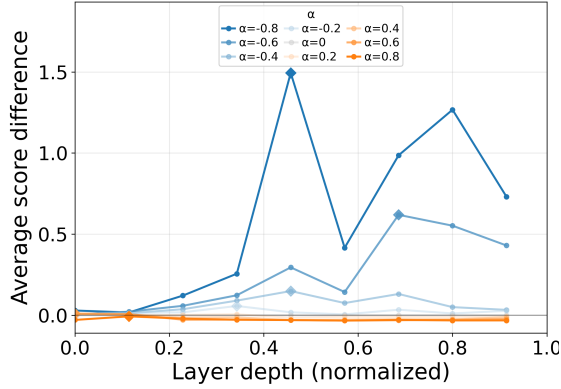
Evolution across training stages. We repeat the steering analysis on Olmo3 7B and 32B at the four training checkpoints, see Figure 6. These results prompt two observations. First, across both model sizes, the highest $\Delta J(\alpha, \ell)$ is nonzero at every stage, including Base. Second, for the Base models, the most responsive layers are scattered without a clear single peak; after SFT the response to the steering becomes more limited to fewer layers that are stable across the subsequent



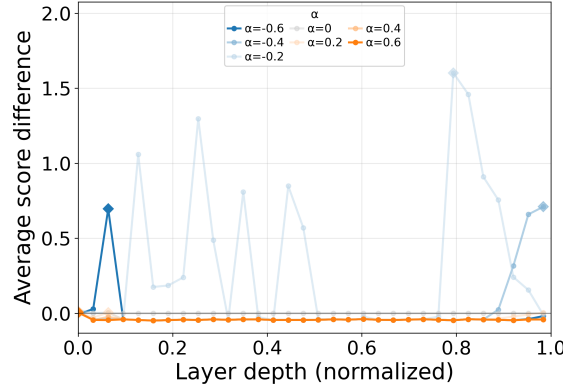
(a) Olmo3-7B



(b) Olmo3-32B



(c) Qwen3-8B



(d) Qwen3-32B

Figure 5: Steering effect across Olmo3 and Qwen3 models. Each line corresponds to a fixed steering strength α ; the y-axis is $\Delta J(\alpha, \ell)$, the change in average judge score relative to the unsteered baseline; the x-axis is the normalized layer depth at which the intervention is applied. Diamonds mark the highest delta at each α . The range of judged layers for each model is as follows: Qwen3-8B: every four layer from 0 to 32. Qwen3-32B: every two layers from 0 to 62. Olmo3-7B: every four layers from 0 to 28. Olmo3-32B: every four layers from 22 to 40. The ranges are inclusive.

training steps. This pattern is consistent with the AUROC and probe-direction-cosine results observed in §3, and again suggests the central role that SFT plays in shaping the representation of evaluation awareness. See Appendix I.2 for more details.

5 Discussion

Detection, expression, and control are not the same signal. Representation, verbalization, and causal influence measurements are related but not equivalent. Across all six models and checkpoints, the probe separates evaluation from deployment prompts with AUROC above 0.7 at the best layer. At the same time, the probe scores show only weak association with the verbalization scores on individual prompts ($|\rho| < 0.19$ and mutual information < 0.04 nats). Nevertheless, steering along the probe direction significantly changes the verbalization behavior in several models. These results suggest that

an internal representation of evaluation awareness does not necessarily predict whether that information will be verbalized, nor how strongly it will be reflected in the model’s outputs. Studying evaluation awareness therefore requires considering representation, verbalization, and intervention as distinct facets of the phenomenon.

Implications for benchmark-based evaluation. Benchmarks and safety evaluations rely on the assumption that behavior observed during evaluation is informative about behavior in deployment. However reasonable, this assumption might break should models condition their behavior on signals indicating they are under evaluation. However, measuring the extent to which a model is “evaluation aware” is challenging and, indeed, our results show that different measurements of evaluation awareness provide different information. A model may internally distinguish evaluation from deployment while rarely verbalizing that distinction (and hence being harder